

Boundary-Localized Genomic Utility and Clinical-Tail Dominance: A Likelihood-Ratio Theory for ZeBRA and Fibrotic Interstitial Lung Disease

Ishanu Chattopadhyay

Division of Biomedical Informatics, University of Kentucky College of Medicine

Abstract—Genomic risk and longitudinal clinical history are distinct information channels for forecasting future disease. We study when a static germline genomic channel can add predictive value once a longitudinal clinical-history score has become strongly informative. The central object is the residual genomic likelihood ratio conditional on clinical history. If this residual genomic evidence is uniformly bounded while sequential clinical evidence can develop an extended likelihood-ratio tail, then two consequences follow. First, genomic information can change a likelihood-ratio decision only inside a bounded neighborhood of the clinical decision threshold; outside that region the clinical evidence is already too strong for bounded genomic evidence to reverse the decision. Second, at sufficiently extreme specificity, the clinical likelihood-ratio test strictly dominates every genomic-only rule whose likelihood ratio is bounded. We also show that any deterministic genomic model inherits the bound and that decision-local benefit can coexist with negligible global AUC change. We evaluate these predictions in a Colorado fibrotic interstitial lung disease (ILD) analysis using ZeBRA, a longitudinal EHR-derived score, and a curated genomic panel containing *MUC5B* rs35705950. Across repeated held-out splits, ZeBRA dominates global discrimination, broad genomic augmentation provides essentially no global FILD/FILA gain after conditioning on ZeBRA, and ZeBRA predicts *MUC5B* carrier status only at chance level. Nevertheless, *MUC5B* retains significant conditional disease information in restricted ZeBRA operating bands, and a two-threshold genomic rescue rule increases held-out sensitivity by up to approximately 4.5 percentage points at essentially matched false-positive rate. The empirical pattern is therefore not global genomic redundancy but regime-dependent complementarity localized near the clinical decision boundary.

Index Terms—ZeBRA, fibrotic interstitial lung disease, idiopathic pulmonary fibrosis, *MUC5B*, genomics, likelihood ratio, Neyman–Pearson lemma, Bayesian inference, decision boundary.

I. MOTIVATING PROBLEM: ZEBRA AND GENOMIC RISK IN FIBROTIC ILD

Fibrotic ILDs, including idiopathic pulmonary fibrosis (IPF), have a strong but non-Mendelian genetic component. The *MUC5B* promoter variant rs35705950 is among the strongest common genetic risk factors for pulmonary fibrosis [2]–[4]. In the original report, the minor allele was present much more frequently among subjects with IPF than among controls [2]. At the same time, disease emergence is accompanied by a growing sequence of clinically observable diagnoses, prescriptions, procedures, and comorbidities. ZeBRA is designed to summarize such routinely observed longitudinal clinical histories into a

future-risk score without requiring laboratory tests, imaging, or questionnaires [1].

Our recent fibrotic ILD analyses ask whether genomic information improves prediction once ZeBRA is known. In 30 repeated held-out splits of the current operational FILD/FILA endpoint, mean AUC is approximately 0.814 for ZeBRA, 0.566 for the selected genomic panel, and 0.799 for a nonlinear combined model. Because that comparison alone does not isolate incremental information, we also performed a paired regularized analysis in which ZeBRA-only and ZeBRA-plus-genomics models were evaluated on the same held-out subjects. For FILD/FILA, the mean paired AUC increment from the complete genomic panel is only +0.00043, the median increment is 0, and the split-wise 2.5th–97.5th percentile range spans approximately -0.0051 to $+0.0041$; average precision decreases and both Brier score and log loss worsen after broad genomic augmentation.

The absence of global gain is not explained by ZeBRA reconstructing the major genomic signal. Among subjects with called rs35705950 genotype, ZeBRA predicts *MUC5B* carrier status at chance level (AUC approximately 0.509). A modest pooled enrichment of carriers in the extreme ZeBRA tail disappears after stratification by FILD/FILA status, showing that the pooled association is consistent with both channels tracking disease rather than with clinical history encoding genotype. Yet residual *MUC5B* disease information is strongly heterogeneous across ZeBRA score space. In the ZeBRA band corresponding approximately to 5–10% control FPR, adding *MUC5B* after ZeBRA yields a likelihood-ratio statistic of 16.55 ($p \approx 4.7 \times 10^{-5}$). A two-threshold rule that consults *MUC5B* only between a stringent upper and lower ZeBRA threshold increases held-out sensitivity by up to approximately 4.5 percentage points while preserving essentially the same realized overall FPR. These observations motivate and correspond to the theory below; they are not taken as proofs of its regularity assumptions.

The apparent paradox is therefore:

A strong genomic risk factor can be biologically important, and can improve decisions for selected individuals, while contributing almost nothing to global discrimination once a strong clinical-history predictor is available.

We show that this behavior is expected under a general Bayesian likelihood-ratio formulation.

II. LATENT-STATE AND INFORMATION-CHANNEL FORMULATION

Let B_t denote the latent biological state at time t , G the germline genomic state, and $E_{0:t}$ the accumulated non-genetic exposure history. The biological state evolves under the influence of G and $E_{0:t}$. For a fixed future horizon $h > 0$, define the disease-risk functional

$$\rho_h(B_t) = \mathbb{P}(T \leq t+h \mid T > t, B_t), \quad (1)$$

where T is the disease-event time. The clinical event sequence $C_{0:t}$ is a noisy observation of the latent biological trajectory $B_{0:t}$. In the idealized Bayesian limit, a clinical-history predictor satisfies

$$Z_h(C_{0:t}) = \mathbb{P}(T \leq t+h \mid C_{0:t}) = \int \rho_h(b) d\mathbb{P}(B_t = b \mid C_{0:t}). \quad (2)$$

Thus ZeBRA need not reconstruct B_t ; it estimates the disease-relevant functional of B_t from the clinical observation channel.

For the theorem it is enough to reduce the problem to binary prediction. Let

$$Y = \mathbf{1}\{T \leq t+h\} \in \{0, 1\}, \quad (3)$$

and write C for the available clinical history. The genomic channel G and clinical channel C are then placed on equal statistical footing through their disease-versus-control likelihood ratios.

III. LIKELIHOOD-RATIO REPRESENTATION

Let $\mathbb{P}_1(\cdot) = \mathbb{P}(\cdot \mid Y = 1)$ and $\mathbb{P}_0(\cdot) = \mathbb{P}(\cdot \mid Y = 0)$. Assume the relevant distributions admit densities with respect to common dominating measures. Define the clinical likelihood ratio

$$\Lambda_C(c) = \frac{p(c \mid Y = 1)}{p(c \mid Y = 0)}, \quad (4)$$

and the residual genomic likelihood ratio conditional on clinical history

$$\Lambda_{G|C}(g; c) = \frac{p(g \mid Y = 1, C = c)}{p(g \mid Y = 0, C = c)}. \quad (5)$$

The full joint likelihood ratio is

$$\Lambda_{C,G}(c, g) = \frac{p(c, g \mid Y = 1)}{p(c, g \mid Y = 0)}. \quad (6)$$

By the probability chain rule,

$$\Lambda_{C,G}(c, g) = \Lambda_C(c) \Lambda_{G|C}(g; c). \quad (7)$$

No conditional-independence assumption is required.

If Z is a calibrated clinical posterior, $Z = \mathbb{P}(Y = 1 \mid C)$, and $\pi = \mathbb{P}(Y = 1)$, Bayes' rule yields

$$\frac{Z}{1-Z} = \frac{\pi}{1-\pi} \Lambda_C(C), \quad (8)$$

so thresholding calibrated ZeBRA is equivalent to thresholding Λ_C . More generally, the same results apply whenever the clinical score is a strictly monotone transformation of Λ_C .

IV. MAIN THEOREM

The key biological assumptions are deliberately weak.

A1 (Uniformly bounded residual genomic evidence). There exist constants $0 < m \leq 1 \leq M < \infty$ such that

$$0 < m \leq \Lambda_{G|C}(g; c) \leq M < \infty \quad (9)$$

for the relevant support of (g, c) , and the marginal genomic likelihood ratio

$$\Lambda_G(g) = \frac{p(g \mid Y = 1)}{p(g \mid Y = 0)} \quad (10)$$

is bounded above by $M_G < \infty$. Complex, non-Mendelian disease architecture motivates finite residual genomic Bayes factors, but does not by itself imply a uniform bound over all conditioned clinical states; boundedness is therefore an explicit regularity assumption.

A2 (Accumulating clinical evidence). Clinical history is sequential, $C_n = (X_1, \dots, X_n)$, and continued diagnoses, prescriptions, procedures, and comorbidities can produce an extended disease-evidence tail. Formally, Λ_C is not bounded above on the relevant sequence of clinical histories: for arbitrarily large K there are events with positive $\mathbb{P}_0$ probability on which $\Lambda_C \geq K$.

Theorem 1 (Boundary-localized genomic utility and clinical-tail dominance). *Under A1–A2:*

(i) Exact factorization. *The optimal joint disease-versus-control likelihood ratio is*

$$\Lambda_{C,G} = \Lambda_C \Lambda_{G|C}. \quad (11)$$

(ii) Boundary localization. *For any likelihood-ratio decision threshold $\lambda > 0$, genomic information can change which side of the threshold an observation occupies only if*

$$\boxed{\frac{\lambda}{M} \leq \Lambda_C(c) \leq \frac{\lambda}{m}}. \quad (12)$$

Equivalently,

$$\boxed{-\log M \leq \log \Lambda_C(c) - \log \lambda \leq -\log m}. \quad (13)$$

Hence the genomic decision-relevant region has fixed width

$$\log(M/m) \quad (14)$$

on the clinical log-likelihood scale, independent of where the decision boundary λ is placed.

(iii) Genomic-only power is bounded. *For any test $\phi(G) \in [0, 1]$ using genomics alone, with false-positive rate $\alpha = \mathbb{E}_0[\phi(G)]$, its true-positive rate satisfies*

$$\boxed{\mathbb{E}_1[\phi(G)] \leq M_G \alpha}. \quad (15)$$

Thus the positive likelihood ratio of any genomic-only decision rule is at most M_G .

(iv) Clinical superiority at extreme specificity. *For any $K > M_G$ with $\alpha_K = \mathbb{P}_0(\Lambda_C \geq K) > 0$, the clinical likelihood-ratio test*

$$\phi_K(C) = \mathbf{1}\{\Lambda_C(C) \geq K\} \quad (16)$$

has

$$\frac{\mathbb{P}_1(\Lambda_C \geq K)}{\mathbb{P}_0(\Lambda_C \geq K)} \geq K > M_G. \quad (17)$$

Therefore, at the same FPR α_K , its power is strictly greater than that of every genomic-only test. Moreover,

$$\alpha_K \leq \frac{1}{K} \rightarrow 0 \quad (18)$$

as $K \rightarrow \infty$. Hence the superiority occurs in the extreme-specificity limit.

(v) Clinical-tail dominance in the combined evidence. Along sequences with $\Lambda_C \rightarrow \infty$,

$$\log \Lambda_{C,G} = \log \Lambda_C + O(1), \quad (19)$$

and therefore

$$\frac{\log \Lambda_{C,G}}{\log \Lambda_C} \rightarrow 1. \quad (20)$$

Thus bounded genomic evidence is asymptotically negligible relative to the accumulated clinical log-likelihood in the extreme clinical-evidence tail.

Proof. Part (i) is the chain rule:

$$\Lambda_{C,G}(c, g) = \frac{p(c, g | Y = 1)}{p(c, g | Y = 0)} \quad (21)$$

$$= \frac{p(c | Y = 1)p(g | Y = 1, c)}{p(c | Y = 0)p(g | Y = 0, c)} \quad (22)$$

$$= \Lambda_C(c)\Lambda_{G|C}(g; c). \quad (23)$$

For part (ii), compare the clinical-only decision $\Lambda_C \geq \lambda$ with the joint decision $\Lambda_C\Lambda_{G|C} \geq \lambda$. If

$$\Lambda_C < \lambda/M, \quad (24)$$

then A1 gives

$$\Lambda_C\Lambda_{G|C} \leq \Lambda_C M < \lambda, \quad (25)$$

so no genomic state can move the observation above the threshold. Conversely, if

$$\Lambda_C > \lambda/m, \quad (26)$$

then

$$\Lambda_C\Lambda_{G|C} \geq \Lambda_C m > \lambda, \quad (27)$$

so no genomic state can move the observation below the threshold. Hence a threshold change is possible only on the interval in (12). Taking logarithms gives (13); its width is

$$(-\log m) - (-\log M) = \log(M/m), \quad (28)$$

which does not depend on λ .

For part (iii), let $\phi(G) \in [0, 1]$. Using the Radon–Nikodym identity under $\mathbb{P}_0$,

$$\mathbb{E}_1[\phi(G)] = \mathbb{E}_0[\Lambda_G(G)\phi(G)] \quad (29)$$

$$\leq M_G \mathbb{E}_0[\phi(G)] \quad (30)$$

$$= M_G \alpha. \quad (31)$$

This proves (15). The result holds for randomized as well as deterministic genomic rules.

For part (iv), define $A_K = \{c : \Lambda_C(c) \geq K\}$. Then

$$\mathbb{P}_1(A_K) = \mathbb{E}_0[\Lambda_C(C)\mathbf{1}\{A_K\}] \quad (32)$$

$$\geq K \mathbb{E}_0[\mathbf{1}\{A_K\}] \quad (33)$$

$$= K \mathbb{P}_0(A_K) \quad (34)$$

$$= K \alpha_K. \quad (35)$$

Therefore the clinical rule has LR+ at least K . If $K > M_G$, any genomic-only test with the same FPR α_K has power at most $M_G \alpha_K < K \alpha_K \leq \mathbb{P}_1(A_K)$. By the Neyman–Pearson lemma [5], thresholding Λ_C is also the most powerful rule among all tests measurable with respect to C at the corresponding size (with the usual boundary randomization if needed).

Finally, because $\mathbb{E}_0[\Lambda_C] = 1$, Markov's inequality gives

$$\alpha_K = \mathbb{P}_0(\Lambda_C \geq K) \leq \frac{\mathbb{E}_0[\Lambda_C]}{K} = \frac{1}{K}, \quad (36)$$

so $\alpha_K \rightarrow 0$ as $K \rightarrow \infty$. This is precisely the high-specificity regime.

For part (v), A1 implies

$$\log m \leq \log \Lambda_{G|C} \leq \log M, \quad (37)$$

so $\log \Lambda_{G|C} = O(1)$. By part (i),

$$\log \Lambda_{C,G} = \log \Lambda_C + \log \Lambda_{G|C} = \log \Lambda_C + O(1). \quad (38)$$

Dividing by $\log \Lambda_C$ and letting $\Lambda_C \rightarrow \infty$ yields (20). $\square$

Corollary 1 (Posterior-risk form for ZeBRA). *If ZeBRA is calibrated so that $Z = \mathbb{P}(Y = 1 | C)$, let $\tau \in (0, 1)$ be any ZeBRA decision threshold and define posterior-odds threshold $\kappa = \tau/(1 - \tau)$. Then genomic information can change the decision only if*

$$\frac{\kappa}{M} \leq \frac{Z}{1 - Z} \leq \frac{\kappa}{m}. \quad (39)$$

Equivalently,

$$-\log M \leq \text{logit}(Z) - \text{logit}(\tau) \leq -\log m. \quad (40)$$

Thus genomic utility is localized around whatever ZeBRA operating boundary is chosen.

Proof. Equation (8) shows that ZeBRA posterior odds differ from Λ_C by the positive constant prior odds $\pi/(1 - \pi)$. Substitution into Theorem 1(ii) yields the result. $\square$

Corollary 2 (Any genomic model inherits the bound). *Let $S = f(G)$ be any measurable genomic predictor, including a polygenic risk score, gradient-boosted model, random forest, neural network, or generative genomic model. If the full genomic likelihood ratio is bounded, then the likelihood ratio of S is also bounded. Therefore no deterministic compression or machine-learning transformation of the same genomic information can evade the finite-evidence and extreme-specificity conclusions of Theorem 1.*

Proof. For a score value s , the score likelihood ratio can be written as a conditional average of the full genomic likelihood ratio under $\mathbb{P}_0$:

$$\Lambda_S(s) = \mathbb{E}_0[\Lambda_G(G) | S = s]. \quad (41)$$

A conditional expectation of a random variable bounded above by M_G is itself bounded above by M_G . The analogous lower bound follows if Λ_G is bounded away from zero. $\square$

Proposition 1 (Local decision gain can coexist with negligible global AUC change). *Let $S(C)$ be a clinical score and $S'(C, G)$ an augmented score. Suppose $S' = S$ except on a measurable region B of the clinical-genomic state space. Then*

$$|\text{AUC}(S') - \text{AUC}(S)| \leq \mathbb{P}_1(B) + \mathbb{P}_0(B) - \mathbb{P}_1(B)\mathbb{P}_0(B) \quad (42)$$

$$\leq \mathbb{P}_1(B) + \mathbb{P}_0(B). \quad (43)$$

Hence an augmented rule can produce a substantial change in decisions within a small boundary region while producing an arbitrarily small change in global AUC as the case and control mass of that region tends to zero.

Proof. AUC is the probability that a randomly drawn case receives a higher score than a randomly drawn control, with the usual half-credit convention for ties. If neither member of a case-control pair lies in B , then $S' = S$ for both members and the pairwise ranking contribution is unchanged. Therefore the AUC contribution can differ only when at least one member of the pair lies in B . The probability of that event is

$$1 - (1 - \mathbb{P}_1(B))(1 - \mathbb{P}_0(B)) \quad (44)$$

$$= \mathbb{P}_1(B) + \mathbb{P}_0(B) - \mathbb{P}_1(B)\mathbb{P}_0(B), \quad (45)$$

which bounds the maximum possible AUC change. $\square$

V. WHY THE CLINICAL CHANNEL CAN HAVE AN EXTENDED TAIL

The theorem does not assume that every later clinical event is informative, nor that disease becomes perfectly observable. It requires only that the clinical channel can accumulate sufficiently strong evidence on some disease trajectories. For a sequential history $C_n = (X_1, \dots, X_n)$, the clinical likelihood ratio factorizes exactly as

$$\Lambda_C(C_n) = \prod_{k=1}^n \frac{p(X_k | Y = 1, X_{<k})}{p(X_k | Y = 0, X_{<k})}, \quad (46)$$

so that

$$\log \Lambda_C(C_n) = \sum_{k=1}^n \log \frac{p(X_k | Y = 1, X_{<k})}{p(X_k | Y = 0, X_{<k})}. \quad (47)$$

This is the standard likelihood-ratio accumulation underlying sequential testing [6]. The genomic state does not acquire new germline variants as the disease approaches, whereas the clinical history can acquire additional diagnoses, procedures, prescriptions, and comorbidities. The difference is therefore not that clinical and genomic predictors obey different probability laws; both enter Bayes' rule as evidence. The difference is that one channel is static and finite while the other is longitudinal and can supply repeated likelihood-ratio factors.

Equation (46) also clarifies the sense in which clinical odds may rise sharply: likelihood evidence adds in log space but multiplies in odds space. No claim of exponential growth

follows from AUC alone. Rather, when a sequence contains repeated positive log-likelihood increments, the posterior odds are the exponential of their accumulated sum.

VI. INTERPRETATION OF THE THEORETICAL RESULT

The theorem and Proposition 1 make four distinct claims that should not be conflated.

A. Global discrimination is not the quantity controlled by the theorem

Theorem 1(ii) is about whether bounded residual genomic evidence can change a threshold decision, not about global ranking. If genomic information matters only in the boundary band (12), then only a restricted fraction of case-control pairs can be reordered. Proposition 1 formalizes the consequence: a meaningful local operating-point improvement can coexist with a negligible global AUC change. Thus “no AUC gain” and “local genomic utility” are mathematically compatible rather than contradictory.

B. Weak ZeBRA-genotype association does not imply zero conditional genomic disease information

The relevant residual quantity is $\Lambda_{G|C}$, not the marginal dependence between the clinical score and genotype. ZeBRA can be nearly uninformative about rs35705950 carrier status while *MUC5B* still changes disease odds conditional on a particular clinical state. The channels can therefore be statistically distinct yet jointly useful near selected boundaries.

C. Decision utility is expected to move with the operating threshold

Corollary 1 shows that the genomic decision-relevant region is defined relative to the chosen ZeBRA threshold. There is no unique genomic “useful band” independent of the decision problem. Changing the tolerated false-positive rate changes the clinical boundary and therefore changes which patients can be reclassified by bounded genomic evidence.

D. Clinical-tail dominance is a high-specificity statement, not an AUC statement

Theorem 1(iii)–(v) concerns the likelihood-ratio tail. A genomic-only rule has LR+ bounded by M_G , whereas a sufficiently extreme clinical likelihood-ratio threshold has LR+ at least K with FPR at most $1/K$. The empirical quantities most relevant to this part of the theory are therefore stringent-specificity likelihood ratios and low-FPR operating behavior, not global AUC by itself.

VII. EMPIRICAL CORRESPONDENCE IN THE COLORADO ILD ANALYSIS

A. Cohort and analysis hierarchy

The current analysis uses the Colorado genomic-clinical cohort assembled in `zebra_comp`. Three stored phenotype fields are evaluated: `FILED` or `FILA` ADJUDICATED, `NONFIBROTIC ILD` ADJUDICATED,

and NON FIBROTIC ILA ADJUDICATED. The historical notebook construction maps Y/y to 1 and N/n to 0, converts the field to numeric, assigns missing target values to 0, and excludes observations without an available 104-week ZeBRA score. The reported endpoint is therefore the notebook-defined operational target; not every target-zero observation is an independently adjudicated negative. This distinction matters especially when interpreting FPR.

The empirical analyses are intentionally hierarchical. Notebook 32 asks whether genomics alone or broad nonlinear ZeBRA+genomics improves global held-out discrimination. Notebook 33 asks the more specific paired question of incremental genomic value conditional on ZeBRA. Separate analyses then test whether ZeBRA reconstructs *MUC5B*, where residual *MUC5B* information lies in ZeBRA score space, and whether a two-threshold genomic rule improves boundary decisions at matched FPR.

B. Global repeated-split discrimination

Across 30 repeated held-out outer splits, ZeBRA is the strongest predictor for all three endpoints. For FILD/FILA, mean AUC is 0.8142 for ZeBRA, 0.5662 for genomics alone, and 0.7988 for the nonlinear combined model. The same qualitative ordering is stronger for the two nonfibrotic endpoints. These results establish the global pattern, but do not by themselves test whether genomics has a small conditional increment after ZeBRA is known.

C. Paired incremental value conditional on ZeBRA

Notebook 33 evaluates ZeBRA-only and ZeBRA-plus-genomics models on the same held-out subjects in 30 repeated splits. For FILD/FILA, the mean paired AUC difference is only +0.00043, the median is 0, and the split-wise 2.5th–97.5th percentile interval is $[-0.00507, 0.00408]$. Average precision decreases, while Brier score and log loss worsen. The conclusion is therefore not merely that one nonlinear combined model underperformed, but that broad genomic augmentation supplies no reproducible global FILD/FILA discrimination gain conditional on ZeBRA.

D. ZeBRA does not reconstruct *MUC5B*

Among subjects with called rs35705950 genotype, ZeBRA predicts carrier status at essentially chance level: pooled AUC 0.5088, explicit FILD/FILA-negative AUC 0.4847, and FILD/FILA-positive AUC 0.4817. Yet carrier prevalence is approximately 39.3% in FILD/FILA-positive subjects versus 19.2% among explicit negatives, and the pooled top 1% ZeBRA tail shows 1.47-fold enrichment. The correct interpretation is therefore that both ZeBRA and *MUC5B* are associated with disease, not that ZeBRA has reconstructed genotype.

E. Residual genomic information is localized

The local-information analysis directly addresses the quantity suggested by Theorem 1(ii). Conditional *MUC5B* evidence varies strongly across ZeBRA-defined control-FPR bands. In the 5–10% band, the likelihood-ratio statistic for adding

MUC5B after ZeBRA is 16.55 ($p = 4.74 \times 10^{-5}$); in the 20–40% band it is 13.69 ($p = 2.15 \times 10^{-4}$). This is not evidence that *MUC5B* globally outperforms ZeBRA. It is evidence that residual genomic disease information is spatially heterogeneous in clinical-score space.

F. Boundary-targeted genomics improves the operating decision

A two-threshold rescue rule consults *MUC5B* only for subjects lying between an upper and a lower ZeBRA threshold. Across all six evaluated threshold pairs, held-out sensitivity increases while realized overall FPR remains essentially unchanged. For the 0.5%/10% pair, sensitivity rises from 0.3587 to 0.4033 (+0.0446), FPR changes from 0.02335 to 0.02310, and LR+ increases from 15.58 to 17.54. This is the most direct empirical counterpart of boundary-localized utility.

G. Low-FPR operating geometry

The low-FPR view is most relevant to the clinical-tail part of the theorem. Across the evaluated matched operating points, rescue shifts sensitivity upward with very small FPR changes and systematically increases LR+. The pointwise envelope in Fig. 6 is descriptive rather than a third classifier; it summarizes the best sensitivity achieved among the evaluated operating rules.

TABLE I
THEORY–EVIDENCE CORRESPONDENCE. THE EMPIRICAL ANALYSES ILLUSTRATE PREDICTED CONSEQUENCES; THEY DO NOT PROVE A1 OR A2.

Theory	Analysis	Empirical meaning
Distinct channels and residual genomic evidence	ZeBRA-to- <i>MUC5B</i> pooled and FILD-stratified analyses	Weak global genomic increment is not explained by ZeBRA reconstructing genotype.
Boundary localization, Theorem 1(ii)	Local ZeBRA- <i>MUC5B</i> information curves and FPR-defined bands	Conditional genomic disease information is heterogeneous across clinical-score space.
Local gain with little global AUC change, Proposition 1	Notebook 32 and notebook 33 plus matched-FPR rescue	Global augmentation yields essentially no FILD AUC gain even though selected boundary decisions improve.
Clinical-tail dominance, Theorem 1(iii)–(v)	Low-FPR sensitivity and LR+ behavior	Stringent clinical operating points can dominate genomic-only evidence even when genomics remains locally useful.

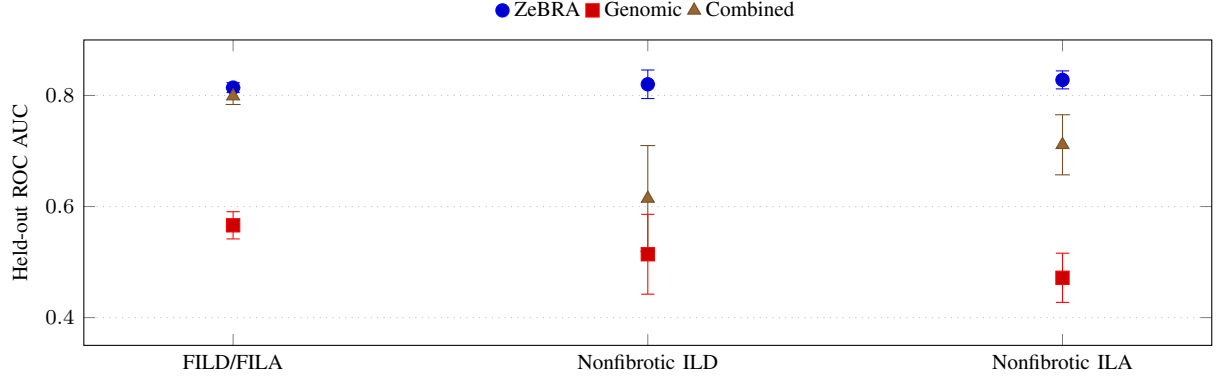


Fig. 1. Global repeated-split discrimination from notebook 32. Points are mean held-out AUC over 30 outer splits; bars indicate one standard deviation. The clinical-history channel dominates globally, and broad nonlinear combination does not improve ZeBRA.

TABLE II
HELD-OUT AUC OVER 30 REPEATED OUTER SPLITS FROM NOTEBOOK 32. VALUES ARE MEAN $\pm$ SD.

Target	ZeBRA	Genomic	Combined
FILD/FILA	0.8142 $\pm$ 0.0089	0.5662 $\pm$ 0.0245	0.7988 $\pm$ 0.0150
Nonfibrotic ILD	0.8202 $\pm$ 0.0257	0.5141 $\pm$ 0.0719	0.6145 $\pm$ 0.0953
Nonfibrotic ILA	0.8281 $\pm$ 0.0162	0.4717 $\pm$ 0.0443	0.7112 $\pm$ 0.0541

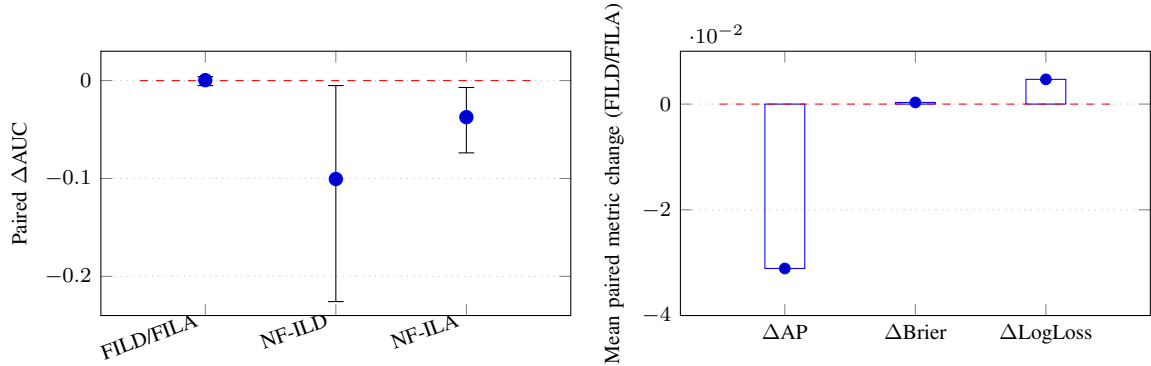


Fig. 2. Paired incremental-value analysis from notebook 33. Left: mean paired Δ AUC from adding the complete genomic panel to ZeBRA, with split-wise 2.5th–97.5th percentile intervals. FILD/FILA is centered essentially at zero. Right: FILD/FILA average precision, Brier score, and log loss changes all move in the unfavorable direction.

TABLE III
LOCAL FILD/FILA INFORMATION IN ZEBRA-DEFINED CONTROL-FPR BANDS.

Band	N	Cases	LRT $G \mid Z$	p	AUC ZeBRA	AUC <i>MUC5B</i>
2–5%	401	26	4.44	0.0352	0.577	0.595
5–10%	652	26	16.55	4.74×10^{-5}	0.367	0.677
10–20%	1,281	30	4.77	0.0289	0.558	0.586
20–40%	2,540	37	13.69	2.15×10^{-4}	0.516	0.635

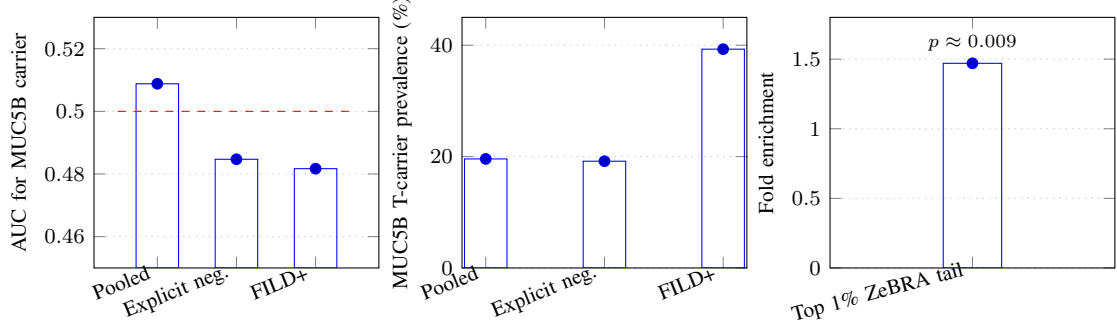


Fig. 3. The weak pooled ZeBRA–*MUC5B* relationship is disease-mediated rather than genotype reconstruction. Left: ZeBRA-to-carrier AUC is at chance in pooled and FILD-stratified analyses. Middle: carrier prevalence is much higher among FILD/FILA-positive subjects. Right: the pooled extreme ZeBRA tail is modestly enriched for carriers.

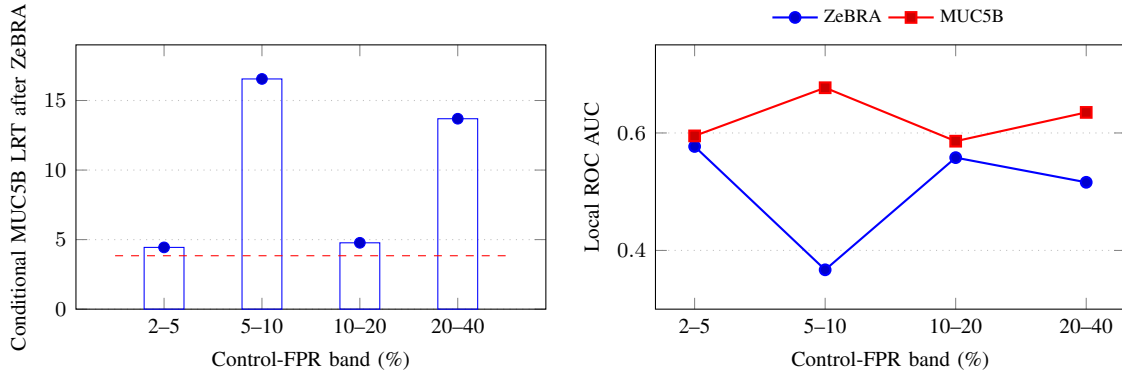


Fig. 4. Localized genomic disease information. Left: conditional *MUC5B* likelihood-ratio statistics after ZeBRA; the dashed line is the $\chi^2_1 = 3.84$ reference. Right: local discrimination varies by operating region, with *MUC5B* stronger than ZeBRA in selected bands.

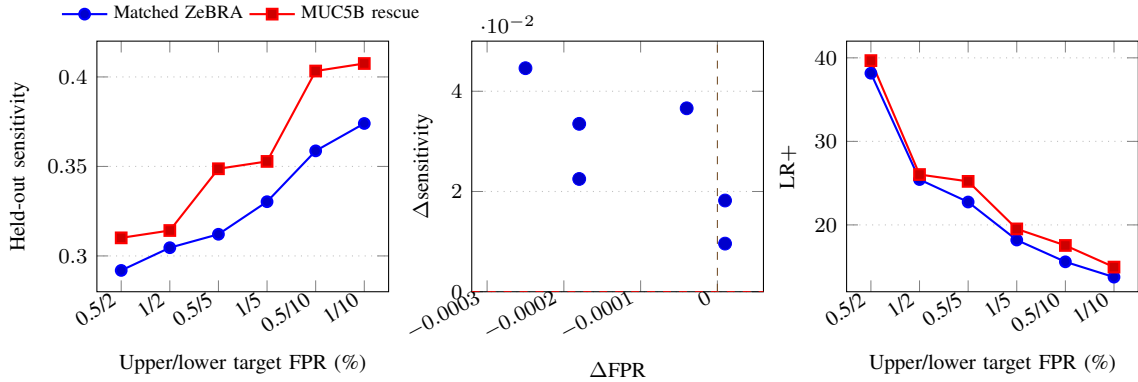


Fig. 5. Matched-FPR boundary rescue. Left: sensitivity of matched ZeBRA and boundary-targeted *MUC5B* rescue. Center: every evaluated pair has positive sensitivity gain while Δ FPR is near zero. Right: LR+ also increases under rescue.

TABLE IV
SELECTED MATCHED-FPR BOUNDARY-RESCUE RESULTS OVER 100 REPEATED SPLITS.

Upper/lower	ZeBRA sens.	Rescue sens.	Δ sens.	ZeBRA FPR	Rescue FPR	ZeBRA LR+	Rescue LR+	Rescue LR–
1%/5%	0.3303	0.3528	+0.0225	0.01844	0.01826	18.19	19.52	0.659
0.5%/5%	0.3121	0.3487	+0.0366	0.01404	0.01400	22.74	25.21	0.661
1%/10%	0.3740	0.4075	+0.0335	0.02753	0.02735	13.75	14.95	0.609
0.5%/10%	0.3587	0.4033	+0.0446	0.02335	0.02310	15.58	17.54	0.611

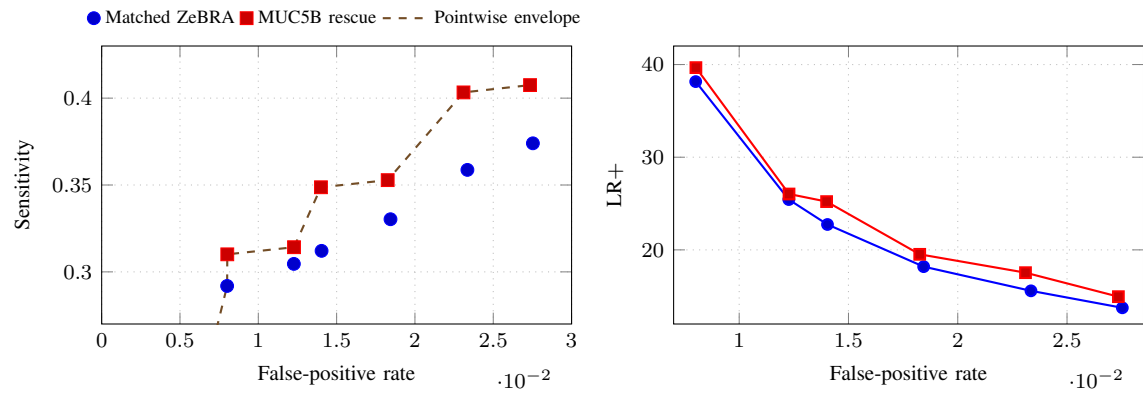


Fig. 6. Low-FPR operating geometry. Boundary-targeted rescue occupies a favorable low-FPR region and increases LR+ without requiring improved global AUC.

VIII. SCOPE AND LIMITATIONS

genomic state through Corollary 2.

Several qualifications are essential. First, A1 is an explicit uniform boundedness assumption; non-Mendelian disease architecture motivates finite genomic evidence but does not prove a uniform bound after conditioning on every clinical state. Second, A2 is also an assumption. Sequential observations make likelihood-ratio accumulation possible, but high AUC alone does not establish an unbounded or sufficiently extended clinical tail. The empirical quantities most relevant to A2 are calibration, score-stratum likelihood ratios, stability of extreme-tail LR+, and fixed-specificity or partial-AUC behavior. Third, Theorem 1(ii) concerns threshold changes, not posterior-probability changes: genomics may still shift posterior risk away from the boundary even when it cannot reverse the decision. Fourth, the operational FILD/FILA target in the historical notebook construction maps missing adjudication values to zero; consequently, the target-zero group should not be described as a fully adjudicated control cohort without further confirmation. Finally, the particular genomic rescue rule remains a modeling result that requires independent or prospectively prespecified validation. The empirical analyses support the predicted geometry but do not validate A1 or A2 by themselves.

A stronger future validation of the theory should therefore test ZeBRA calibration, convert score strata into empirical likelihood ratios, quantify uncertainty in extreme-tail LR+, evaluate prespecified fixed-specificity thresholds, and replicate boundary-localized genomic rescue in an external cohort.

IX. CONCLUSION

The central result is not that clinical history “contains” genomics, nor that genomics is globally useless. The two modalities are distinct evidence channels. Bayes’ rule factorizes their joint evidence into a clinical likelihood ratio and a residual genomic likelihood ratio conditional on clinical state. If the residual genomic factor is bounded while sequential clinical evidence can extend into a strong likelihood-ratio tail, then genomic information can reverse a clinical threshold decision only inside a bounded neighborhood of that threshold, and sufficiently extreme clinical evidence must eventually dominate every bounded genomic-only rule. Any deterministic genomic predictor inherits the same finite-evidence constraint, while Proposition 1 explains why clinically meaningful boundary reclassification can coexist with almost no change in global AUC.

The current Colorado results match this qualitative geometry. ZeBRA dominates global discrimination; broad genomic augmentation provides essentially zero incremental FILD/FILA AUC after conditioning on ZeBRA; ZeBRA does not reconstruct *MUC5B*; residual *MUC5B* disease information is concentrated in selected ZeBRA operating bands; and a boundary-targeted rescue rule improves sensitivity and LR+ at essentially matched overall FPR. The resulting picture is one of *regime-dependent complementarity*: genomics is most useful where the clinical evidence is close enough to the action threshold that a finite residual Bayes factor can still change the decision, whereas longitudinal clinical evidence dominates in the extreme tail.

The theory is a statement about information channels, not about a particular machine-learning architecture. It does not require independence between genomics and clinical history; dependence is retained explicitly through $\Lambda_{G|C}$. It also applies to any deterministic genomic score derived from the measured

REFERENCES

- [1] D. Onishchenko, J. A. Mastrianni, and I. Chattopadhyay, "Passive early screening for Alzheimer's disease and related dementias using EHR comorbidity patterns," *npj Digital Medicine*, vol. 9, 2026, doi: 10.1038/s41746-026-02954-2.
- [2] M. A. Seibold *et al.*, "A common *MUC5B* promoter polymorphism and pulmonary fibrosis," *New England Journal of Medicine*, vol. 364, no. 16, pp. 1503–1512, 2011, doi: 10.1056/NEJMoa1013660.
- [3] T. E. Fingerlin *et al.*, "Genome-wide association study identifies multiple susceptibility loci for pulmonary fibrosis," *Nature Genetics*, vol. 45, no. 6, pp. 613–620, 2013, doi: 10.1038/ng.2609.
- [4] R. J. Allen *et al.*, "Genome-wide association study of susceptibility to idiopathic pulmonary fibrosis," *American Journal of Respiratory and Critical Care Medicine*, vol. 201, no. 5, pp. 564–574, 2020, doi: 10.1164/rccm.201905-1017OC.
- [5] J. Neyman and E. S. Pearson, "On the problem of the most efficient tests of statistical hypotheses," *Philosophical Transactions of the Royal Society of London. Series A*, vol. 231, pp. 289–337, 1933, doi: 10.1098/rsta.1933.0009.
- [6] A. Wald, "Sequential tests of statistical hypotheses," *The Annals of Mathematical Statistics*, vol. 16, no. 2, pp. 117–186, 1945.
- [7] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Hoboken, NJ, USA: Wiley, 2006.